# Jiacong (Fred) Bao

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## Education

**Robert H. Smith School of Business | University of Maryland** — College Park, MD
*Ph.D. Operations Management/Management Science (Minor: Economics)* — *Expected May 2027*

Dissertation: *"Responsible Operations: Essays on Policy, Healthcare, and Supply Chain Accountability"*
Committee: Michael Fu (Chair), Ashish Kabra (Co-Chair), Wedad Elmaghraby, John Silberholz, Anna Alberini

**Olin Business School | Washington University in Saint Louis** — Saint Louis, MO
*M.S. Supply Chain Management* — *2021*

**Southwestern University of Finance and Economics** — Sichuan, China
*B.S. Economics, International Economics and Trade* — *2019*

## Research Interests

Empirical Operations Management, Socially Responsible Operations, Healthcare Operations, Causal Inference, Data-Driven Operations.

## Research

**1. Child Labor Violations and Supply Chain Restructuring in the United States.**
**(Job Market Paper)**
*(with Wedad Elmaghraby and Ashish Kabra.) In preparation for submission to Management Science.*

- **Short Abstract:** Examines how child labor violations at tier-1 suppliers reshape buyers' supply chains and subsequent exposure to child labor risk. The study shows that violations trigger broader supplier restructuring and that terminating violating suppliers reduces future exposure to child labor violations. Yet despite termination's effectiveness, firms with fewer viable alternatives are less likely to terminate violating suppliers, highlighting how supplier market accessibility constrains buyer responses.

**2. The Effect of State-Mandated Employment Certificates on Child Labor Violations in the United States.**
*(with Ashish Kabra.) Accepted for publication at Production and Operations Management (POM).*

- **Short Abstract:** Shows that state requirements of minor Employment Certificates reduce child labor violations, informing ongoing state debates over weakening these requirements.
- Media Mentions: Bloomberg Law, Route Fifty, Economic Policy Institute (1, 2, 3)
- Policy Discussions: CSG Midwest Policy Brief, The Guardian, The Century Foundation
- Legislation: Cited in Arkansas HB 1731; referenced in committee testimony for Michigan HB 5727 and Ohio SB 455
- 2025 POMS CSO Best Student Paper Award–Second Place

**3. When Capacity Constrains Care: ICU Capacity Constraints in Kidney Offer Acceptance Decisions.**
*(with Qiwei Ma, Yunzhe Qiu, Naoru Koizumi, Meng-Hao Li, and Michael Fu.) Major revision at Manufacturing & Service Operations Management (M&SOM).*

- **Short Abstract:** Shows that critical care capacity (ICU) constrains hospitals' acceptance of deceased donor kidney offers, potentially leading to the discard of life-saving organs.
- **Funding:** Supported in part by the National Science Foundation (NSF) under Grants IIS-2123683 and IIS-2123684.

**4. Optimal Effort Under Max-Operator Pay-For-Performance.**
*(with John Silberholz and Michael Fu.) Work in progress.*

- **Short Abstract:** Using a principal-agent framework, we characterize transplant centers' optimal effort under CMS's IOTA scoring rule across ability levels, highlighting how the scoring design may create unintended effects.

**5. Staffing Agency Outsourcing and Child Labor Violations in the United States.**
*(with Ashish Kabra.) Work in progress.*

- **Short Abstract:** Examines how staffing agencies connect to child labor violations as firms increasingly outsource hiring. Informs regulators on targeted enforcement and joint-employer liability for hiring through intermediaries.

**6. Granular Prediction of Child Labor Violation Risk Across the US.**
*(with Wedad Elmaghraby and Ashish Kabra.) Work in progress.*

- **Short Abstract:** Combines diverse data sources to predict granular child labor violation risk across the US, identify the incentives behind it, and inform targeted enforcement.

*Research in Other Disciplines*

**7. Hospital Performance Impact on Deceased Donor Kidney Offer Acceptance.**
*(with Naoru Koizumi, Meng-Hao Li, and Michael Fu.) Under review at Frontiers in Nephrology, Kidney Transplantation section.*

- **Short Abstract:** Shows that transplant centers flagged as performance-deficient under federal oversight are less likely to accept deceased donor kidney offers, suggesting potential unintended consequences.

## Teaching

**BMGT230 Business Statistics** Summer 2026
*Instructor | Course Evaluation: 3.42/4* *College Park, MD*

**BMGT230 Business Statistics** Summer 2025
*Instructor | Course Evaluation: 3.5/4* *College Park, MD*

## Selected Talks

**2026:** INFORMS Annual Meeting–Job Market Showcase; HOPE Workshop; MSOM Annual Meeting; POMS Annual Meeting†; WEOM (presented by coauthor)

**2025:** INFORMS Annual Meeting†; MSOM Annual Meeting–Sustainable SIG; POMS Annual Meeting

**2024:** INFORMS Annual Meeting; MSOM Annual Meeting; POMS Annual Meeting; Early-Career Sustainable Operations Workshop

**2023:** INFORMS Annual Meeting; POMS Annual Meeting

†Session Chair

## Awards

2026 Michael Ball Endowed Ph.D. Fellowship in Business | Robert H. Smith School of Business, University of Maryland

2026 Long Jiang Graduate Student Fellowship | Robert H. Smith School of Business, University of Maryland

2025 POMS College of Sustainable Operations Best Student Paper Award–Second Place

2021 Outstanding Supply Chain Management Student Award | Washington University in Saint Louis

2021 Project of the Year, Boeing Center for Supply Chain Innovation | Washington University in Saint Louis

2020 Unilever Industrial Automation Hackathon–National Round Winner

2019 ASCM Case Competition–School Round Winner

2016/2018 Outstanding Undergraduate Scholarship

2017/2018 Outstanding Volunteer Award

## Professional Service

**Reviewer**
*INFORMS Journal on Computing, Transplantation*

**Session Chair**
*2025 INFORMS Annual Meeting, 2026 POMS Annual Meeting*

**INFORMS University of Maryland Student Chapter** 2024 – Present
*Treasurer* *College Park, MD*

## Other Experience

**Seed Advisor | Bayer Crop Science** — Feb 2021 – May 2022
*Research Assistant* — *Saint Louis, MO*

**Supply Chain Analytics Capstone | Washington University in Saint Louis** — Spring 2021
*Teaching Assistant* — *Saint Louis, MO*

**Boeing Center for Supply Chain Innovation | Washington University in Saint Louis** — Sep 2020 – Mar 2022
*Analytical Consultant* — *Saint Louis, MO*

**HybridHELIX Consulting** — Jun 2020 – Aug 2020
*Intelligence Management Analyst* — *Albuquerque, NM*

**Farmers Insurance** — May 2020 – Jun 2020
*Student Consultant* — *Saint Louis, MO*

**French National Railways** — Sep 2018 – Dec 2018
*Logistics Intern* — *Shanghai, China*

**Johnson & Johnson** — Jan 2018 – May 2018
*Supply Chain Management Intern* — *Shanghai, China*

**SWUFE Youth Volunteer Association** — Nov 2017 – Jan 2018
*Volunteer Teacher* — *Sichuan, China*

## Non-Academic Projects

**Bayer Crop Science** — 2021 – 2022
*Research Assistant to Lingxiu Dong and Durai Sundaramoorthi* — *Saint Louis, MO*

- Predict-then-optimize framework for soybean yield portfolio optimization (2022)

**Boeing Center for Supply Chain Innovation** — 2020 – 2022
*Analytical Consultant* — *Saint Louis, MO*

- Anheuser-Busch InBev: trailer pool sizing via simulation (2022); dedicated fleet routing optimization (2020)
- Inbound supply chain strategy under disruption (2022)
- West Pharma: lead time optimization for COVID vaccine components (2021)
- WashU: queueing model for COVID-safe class scheduling optimization (2021)
- MilliporeSigma: inventory optimization via under-utilized SKUs (2021)
- Belden: ML pricing and customer behavior analysis (2020)
- Express Scripts: ML demand forecasting for pharmaceutical SKUs (2020)

## Skills

**Software:** Python, R, Stata, SQL
**Languages:** English, Mandarin Chinese

## References

**Michael C. Fu**
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**Ashish Kabra**
*Assistant Professor, Nanyang Business School, NTU | ashish.kabra@ntu.edu.sg*

**Wedad Elmaghraby**
*Dean's Chair of Operations Management, Robert H. Smith School of Business, University of Maryland | welmaghr@umd.edu*

**Naoru Koizumi**
*Professor, Schar School of Policy and Government, George Mason University | nkoizumi@gmu.edu*

**John Silberholz**
*Associate Professor of Business Analytics, Robert H. Smith School of Business, University of Maryland | josilber@umd.edu*